

Chapter 4 *UNFINISHED ROUGH DRAFT*

Potential Flow over Airfoils: Panel Methods

4.1. Airfoil Geometry and Nomenclature

An airfoil is any two-dimensional cross-section of a wing that lies in a plane perpendicular to the spanwise coordinate. An airfoil section is completely defined by the geometric shape of its surface. However, the aerodynamic properties of an airfoil section are most profoundly affected by the shape of its centerline. This centerline is midway between the upper and lower surfaces of the airfoil and is called the camber line. If the airfoil is not symmetric, the camber line is not a straight line but rather a planar curve.

Because the shape of the camber line is such an important factor in airfoil design, it is critical that the student understands exactly how the camber line is defined. In addition, there are several other designations that will be used throughout this and following chapters when referring to the geometric attributes of airfoil sections. The student should understand and remember the following nomenclature as it applies to airfoil geometry such as that shown in Fig. 4.1.1.

The **camber line** is the locus of points midway between the upper and lower surfaces of an airfoil section as **measured perpendicular to the camber line itself**.

The **leading edge** is the most forward point on the camber line.

The **trailing edge** is the most rearward point on the camber line.

The **chord line** is a straight line connecting the leading edge and the trailing edge.

The **chord length**, often referred to simply as the **chord**, is the distance between the leading edge and the trailing edge as measured along the chord line.

The **local camber**, at any point along the chord line, is the distance between the chord line and the camber line as **measured perpendicular to the chord line**.

The **maximum camber**, often referred to simply as the **camber**, is the maximum distance between the chord line and the camber line as measured perpendicular to the chord line.

The **local thickness**, at any point along the chord line, is the distance between the upper and lower surfaces as **measured perpendicular to the camber line**.

The **maximum thickness**, often referred to simply as the **thickness**, is the maximum distance between the upper and lower surfaces as **measured perpendicular to the camber line**.

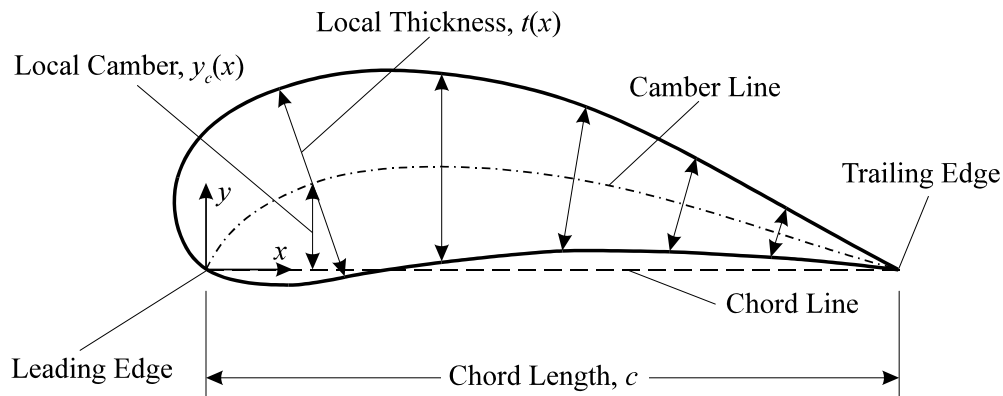


Figure 4.1.1. Airfoil coordinates and nomenclature.

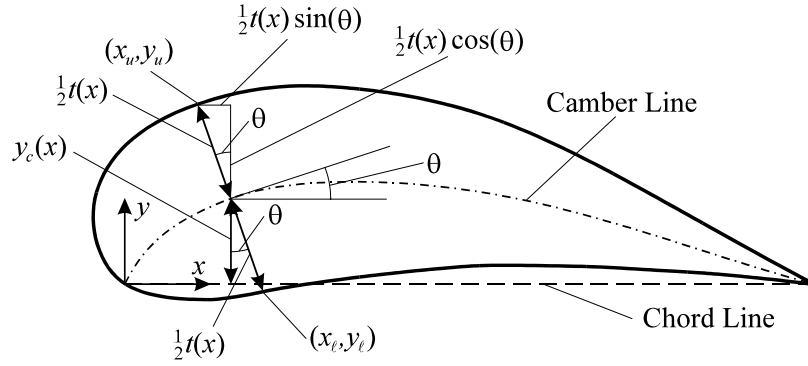


Figure 4.1.2. Generation of airfoil surface geometry from camber line geometry and thickness distribution.

As can be seen from Fig. 4.1.1, the leading edge is not necessarily the most forward point on the surface of an airfoil. For this reason, it is difficult to determine the shape of the camber line from knowledge of the surface geometry alone. If we know only coordinates for every point on the airfoil surface, the location of the leading edge is not explicitly known. Since the location of the leading edge is unknown, both the chord line and the chord length are also unknown. **Because the camber line is defined to be midway between the upper and lower surfaces as measured perpendicular to the camber line itself, neither the camber line nor the chord line can be found explicitly from knowledge of the surface coordinates alone. Thus, even though there is a unique relationship between camber line geometry, thickness distribution, and surface geometry, it becomes a trial and error process to obtain camber line geometry from airfoil surface geometry.**

It is much more convenient to define an airfoil's surface geometry from a known camber line geometry and thickness distribution. If the camber line geometry, $y_c(x)$, and the thickness distribution, $t(x)$, are known, the airfoil surface geometry can be explicitly determined. For any given location on the chord line, x , from Fig. 4.1.2 we see that the x - and y -coordinates of upper and lower surfaces are given by

$$x_u(x) = x - \frac{1}{2}t(x)\sin\theta \quad (4.1.1)$$

$$y_u(x) = y_c(x) + \frac{1}{2}t(x)\cos\theta \quad (4.1.2)$$

$$x_l(x) = x + \frac{1}{2}t(x)\sin\theta \quad (4.1.3)$$

$$y_l(x) = y_c(x) - \frac{1}{2}t(x)\cos\theta \quad (4.1.4)$$

where θ is related to the slope of the camber line through the relation

$$\tan\theta = \frac{dy_c}{dx} \quad (4.1.5)$$

The derivative of y_c with respect to x can be evaluated directly from the known camber line geometry. Since θ is always between $-\pi$ and π , we can use the trigonometric relations

$$\sin\theta = \frac{\tan\theta}{\sqrt{1+\tan^2\theta}} \quad (4.1.6)$$

and

$$\cos\theta = \frac{1}{\sqrt{1+\tan^2\theta}} \quad (4.1.7)$$

Using Eqs. (4.1.5) through (4.1.7) in Eqs. (4.1.1) through (4.1.4), the **upper and lower surface coordinates** for an airfoil can be obtained explicitly from the camber line geometry, $y_c(x)$, and the thickness distribution, $t(x)$,

$$x_u(x) = x - \frac{t(x)}{2\sqrt{1+(dy_c/dx)^2}} \frac{dy_c}{dx} \quad (4.1.8)$$

$$y_u(x) = y_c(x) + \frac{t(x)}{2\sqrt{1+(dy_c/dx)^2}} \quad (4.1.9)$$

$$x_l(x) = x + \frac{t(x)}{2\sqrt{1+(dy_c/dx)^2}} \frac{dy_c}{dx} \quad (4.1.10)$$

$$y_l(x) = y_c(x) - \frac{t(x)}{2\sqrt{1+(dy_c/dx)^2}} \quad (4.1.11)$$

We now see that the shape of an airfoil is completely defined by the shape of its camber line and its thickness distribution. Thus, the aerodynamic characteristics of an airfoil section are also defined by its camber line geometry and thickness distribution. The lift that an airfoil produces at zero angle of attack is determined almost entirely by the camber line geometry. Cambered airfoils in subsonic flow generate lift at zero angle of attack, while symmetric airfoils produce no lift at zero angle of attack. The shape of the camber line also strongly influences the maximum value of airfoil section lift coefficient that can be attained prior to stall.

The thickness distribution also affects the stall characteristics of an airfoil section. The maximum velocity to which the fluid is accelerated, as it flows around an airfoil section, depends on section thickness. The thicker the section, the more the fluid must be accelerated over the leading portion of the airfoil and the more it must be decelerated over the trailing portion. Since pressure is related to fluid velocity, pressure gradient is related to fluid acceleration. Rapidly decelerating flows result in large adverse pressure gradients that promote boundary layer separation and stall. Modifying the section thickness distribution can significantly alter the surface pressure distribution and thus change the stall characteristics of an airfoil section.

The thickness distributions that have been used in subsonic airfoils are widely varied and the possibilities are almost limitless. However, there are six specific features that can be associated with an airfoil thickness distribution and it is these six fundamental features that most directly affect the aerodynamic characteristics of an airfoil section. These features are:

The **leading edge radius**, r_{le} , defined to be the radius of a circle, centered on a line tangent to the camber line at the leading edge, and having the same curvature as the upper and lower surfaces at the leading edge.

The **maximum thickness**, t_m , defined to be the maximum distance between the upper and lower surfaces as measured perpendicular to the camber line.

The **position of maximum thickness**, x_{mt} , defined to be the distance from the leading edge to the point of maximum thickness as measured along the chord line.

The **radius of curvature at maximum thickness**, r_{mt} , defined to be the radius of a circle having the same curvature as the thickness line at the point of maximum thickness.

The **trailing edge thickness**, t_{te} , is either zero or very small, usually on the order of one or two percent of the maximum thickness or less.

The **trailing edge angle**, θ_{te} , defined to be the angle between the upper surface and the lower surface at the trailing edge.

The leading edge of airfoils used for subsonic applications is always rounded and the curvature should be continuous between the upper and lower surfaces, across the leading edge. Recall from calculus and analytic geometry that, for any planar curve where y can be expressed as a function of x , the radius of curvature at any point along the curve can be written as

$$r_c(x) = \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2} \left/ \left| \frac{d^2y}{dx^2} \right| \right. \quad (4.1.12)$$

For a symmetric airfoil section, $y_c = 0$ and the y -coordinates of the upper and lower surfaces are functions only of the thickness distribution. From Eqs. (4.1.9) and (4.1.11), the surface coordinates for a symmetric airfoil are given by

$$y = \pm \frac{1}{2} t(x) \quad (4.1.13)$$

Thus, the radius of curvature for the thickness distribution can be expressed as a function of the axial coordinate,

$$r_c(x) = \left[4 + \left(\frac{dt}{dx} \right)^2 \right]^{3/2} \left/ 4 \left| \frac{d^2t}{dx^2} \right| \right. \quad (4.1.14)$$

The leading edge radius is obtained by applying Eq. (4.1.14) at $x = 0$ and the radius of curvature at maximum thickness is found by evaluating Eq. (4.1.14) for $x = x_{mt}$. The expression for radius of curvature is somewhat simplified when applied at the point of maximum thickness because the change in thickness with respect to x is zero at that point.

To examine how we could force the surface curvature to be continuous across the leading edge of an airfoil section, consider the cross-section of a circular cylinder having radius R and with the leading edge located at the origin. This geometry is defined by the equation

$$(x - R)^2 + y^2 = R^2 \quad (4.1.15)$$

or

$$y(x) = \pm \sqrt{(2R - x)x} \quad (4.1.16)$$

As x approaches zero, $(2R - x)$ approaches $2R$. So, near the leading edge, a circular cross-section has a thickness distribution that is proportional to the square root of x

$$t(x) = (2\sqrt{2R})\sqrt{x} \quad (4.1.17)$$

Thus, to obtain a circular shape near the leading edge of an airfoil section, the thickness distribution must approach some constant multiplied by the square root of x , in the limit as x approaches zero. Most subsonic airfoil sections have a thickness distribution of this type. A fairly general thickness distribution that combines this characteristic with considerable flexibility is given by

$$t(x) = \begin{cases} a_0 \sqrt{\frac{x}{c}} + a_1 \left(\frac{x}{c} \right) + a_2 \left(\frac{x}{c} \right)^2 + a_3 \left(\frac{x}{c} \right)^3, & x \leq x_{mt} \\ b_0 + b_1 \left(1 - \frac{x}{c} \right) + b_2 \left(1 - \frac{x}{c} \right)^2 + b_3 \left(1 - \frac{x}{c} \right)^3 + b_4 \left(1 - \frac{x}{c} \right)^4, & x \geq x_{mt} \end{cases} \quad (4.1.18)$$

The nine coefficients in Eq. (4.1.18) can be determined by specifying the leading edge radius, the maximum thickness, the position of maximum thickness, the radius of curvature at maximum thickness, the trailing edge thickness, and the trailing edge angle. The four constants a_0 , a_1 , a_2 , and a_3 are determined from

$$r_c(0) = r_{le} \quad (4.1.19)$$

$$\frac{dt}{dx}(x_{mt}) = 0 \quad (4.1.20)$$

$$t(x_{mt}) = t_m \quad (4.1.21)$$

$$r_c(x_{mt}) = r_{mt} \quad (4.1.22)$$

The five constants b_0 , b_1 , b_2 , b_3 , and b_4 are determined from

$$\frac{dt}{dx}(x_{mt}) = 0 \quad (4.1.23)$$

$$t(x_{mt}) = t_m \quad (4.1.24)$$

$$r_c(x_{mt}) = r_{mt} \quad (4.1.25)$$

$$t(c) = t_{te} \quad (4.1.26)$$

$$2 \tan^{-1} \left(-\frac{1}{2} \frac{dt}{dx}(c) \right) = \theta_{te} \quad (4.1.27)$$

The explicit method of combining a camber line geometry and a thickness distribution according to Eqs. (4.1.8) through (4.1.11), is sufficiently flexible to permit the generation of any imaginable airfoil shape. Many families of airfoil sections have been developed using this approach. Often the numbering system used for a family of airfoils specifies the section geometry by specifying certain parameters associated with the camber line shape and the thickness distribution. For example, the numbering system for the well-known NACA 4-digit series is based on section geometry. The first digit indicates the maximum camber in percent of chord. The second digit indicates the distance from the leading edge to the point of maximum camber in tenths of the chord. The last two digits indicate the maximum thickness in percent of chord. Thus, a NACA 2412 airfoil section has 2 percent maximum camber located at 40 percent of chord from the leading edge and has a maximum thickness of 12 percent of chord. For a NACA 4-digit airfoil section the leading edge radius, the radius of curvature at maximum thickness, the trailing edge thickness, and the trailing edge angle are all fixed for a given value of maximum thickness and chord.

The section geometry for a **NACA 4-digit series airfoil** is completely fixed by the maximum camber, the location of maximum camber, the maximum thickness, and the chord length. The **camber line** is given by

$$y_c(x) = \begin{cases} y_{mc} \left[2 \left(\frac{x}{x_{mc}} \right) - \left(\frac{x}{x_{mc}} \right)^2 \right], & 0 \leq x \leq x_{mc} \\ y_{mc} \left[2 \left(\frac{c-x}{c-x_{mc}} \right) - \left(\frac{c-x}{c-x_{mc}} \right)^2 \right], & x_{mc} \leq x \leq c \end{cases} \quad (4.1.28)$$

where y_{mc} is the maximum camber and x_{mc} is the position of maximum camber. The **thickness** about the camber line, measured perpendicular to the camber line itself, is given by

$$t(x) = t_m \left[2.969 \sqrt{\frac{x}{c}} - 1.260 \left(\frac{x}{c} \right) - 3.516 \left(\frac{x}{c} \right)^2 + 2.843 \left(\frac{x}{c} \right)^3 - 1.015 \left(\frac{x}{c} \right)^4 \right] \quad (4.1.29)$$

where t_m is the maximum thickness. For a more detailed discussion of subsonic airfoil geometry, including a summary of airfoil section data, see Abbott and Von Doenhoff (1949).

Example 4.1.1. For the NACA 2412 airfoil section, determine x/c and y/c for the camber line, the upper surface, and the lower surface at chordwise positions of $x/c = 0.10$.

Solution. The NACA 2412 airfoil section has 2% maximum camber located at 40% chord. We wish to determine the dimensionless coordinates of the camber line at 10% chord. Thus, we use Eq. (4.1.28) for $0 \leq x \leq x_{mc}$, which gives

$$\frac{x_c}{c} = \frac{x}{c} = 0.10$$

$$\frac{y_c}{c} = \frac{y_{mc}}{c} \left[2 \left(\frac{x/c}{x_{mc}/c} \right) - \left(\frac{x/c}{x_{mc}/c} \right)^2 \right] = 0.02 \left[2 \left(\frac{0.10}{0.40} \right) - \left(\frac{0.10}{0.40} \right)^2 \right] = 0.00875$$

Similarly, from Eq. (4.1.29), the dimensionless thickness at 10% chord is

$$\begin{aligned} \frac{t}{c} &= \frac{t_m}{c} \left[2.969 \sqrt{\frac{x}{c}} - 1.260 \left(\frac{x}{c} \right) - 3.516 \left(\frac{x}{c} \right)^2 + 2.843 \left(\frac{x}{c} \right)^3 - 1.015 \left(\frac{x}{c} \right)^4 \right] \\ &= 0.12 \left[2.969 \sqrt{0.1} - 1.260(0.1) - 3.516(0.1)^2 + 2.843(0.1)^3 - 1.015(0.1)^4 \right] \\ &= 0.0936554 \end{aligned}$$

From Eq. (4.1.28), the camber line slope at 10% chord is

$$\frac{dy_c}{dx} = y_{mc} \left[2 \left(\frac{1}{x_{mc}} \right) - 2 \left(\frac{x}{x_{mc}^2} \right) \right] = 2 \frac{y_{mc}/c}{x_{mc}/c} \left[1 - \left(\frac{x/c}{x_{mc}/c} \right) \right] = 2 \frac{0.02}{0.40} \left[1 - \left(\frac{0.10}{0.40} \right) \right] = 0.075$$

Using Eqs. (4.1.8) through (4.1.11), the dimensionless coordinates of the upper and lower surfaces at 10% chord are

$$\frac{x_u}{c} = \frac{x}{c} - \frac{t/c}{2\sqrt{1+(dy_c/dx)^2}} \frac{dy_c}{dx} = 0.10 - \frac{0.0936554}{2\sqrt{1+(0.075)^2}} 0.075 = 0.096498$$

$$\frac{y_u}{c} = \frac{y_c}{c} + \frac{t/c}{2\sqrt{1+(dy_c/dx)^2}} = 0.00875 + \frac{0.0936554}{2\sqrt{1+(0.075)^2}} = 0.055447$$

$$\frac{x_\ell}{c} = \frac{x}{c} + \frac{t/c}{2\sqrt{1+(dy_c/dx)^2}} \frac{dy_c}{dx} = 0.10 + \frac{0.0936554}{2\sqrt{1+(0.075)^2}} 0.075 = 0.103502$$

$$\frac{y_\ell}{c} = \frac{y_c}{c} - \frac{t/c}{2\sqrt{1+(dy_c/dx)^2}} = 0.00875 - \frac{0.0936554}{2\sqrt{1+(0.075)^2}} = -0.037947$$

Example 4.1.2. For a symmetric NACA 4-digit series airfoil section obtain an equation for, (a) the leading edge radius, (b) the radius of curvature at maximum thickness, (c) the trailing edge thickness, and (d) the trailing edge angle. Evaluate these parameters for the NACA 0012.

Solution. Differentiating the thickness distribution for the NACA 4-digit series airfoil section, as given by Eq. (4.1.29), we have

$$\frac{dt}{dx} = \frac{t_m}{c} \left[\frac{2.969}{2} \left(\frac{x}{c} \right)^{-1/2} - 1.260 - 2(3.516) \left(\frac{x}{c} \right) + 3(2.843) \left(\frac{x}{c} \right)^2 - 4(1.015) \left(\frac{x}{c} \right)^3 \right] \quad (4.1.30)$$

and

$$\frac{d^2t}{dx^2} = \frac{t_m}{c^2} \left[-\frac{2.969}{4} \left(\frac{x}{c} \right)^{-3/2} - 2(3.516) + 6(2.843) \left(\frac{x}{c} \right) - 12(1.015) \left(\frac{x}{c} \right)^2 \right] \quad (4.1.31)$$

Using Eqs. (4.1.30) and (4.1.31) in Eq. (4.1.14), the radius of curvature at the leading edge is

$$\begin{aligned} r_c(0) &= \lim_{x \rightarrow 0} \left\{ \left[4 + \left(\frac{dt}{dx} \right)^2 \right]^{3/2} / 4 \left| \frac{d^2t}{dx^2} \right| \right\} \\ &= \left\{ \frac{t_m}{c} \left[\frac{2.969}{2} \left(\frac{x}{c} \right)^{-1/2} \right] \right\}^3 / 4 \frac{t_m}{c^2} \left[\frac{2.969}{4} \left(\frac{x}{c} \right)^{-3/2} \right] = \frac{2.969^2 t_m^2}{8c} \end{aligned}$$

or

$$\frac{r_{le}}{c} = \frac{2.969^2}{8} \left(\frac{t_m}{c} \right)^2 = 1.10187 \left(\frac{t_m}{c} \right)^2$$

The point of maximum thickness is the point where the change in thickness with respect to x is zero. Thus, using Eq. (4.1.30), the point of maximum thickness is found from

$$1.4845 \left(\frac{x_{mt}}{c} \right)^{-1/2} - 1.260 - 7.032 \left(\frac{x_{mt}}{c} \right) + 8.529 \left(\frac{x_{mt}}{c} \right)^2 - 4.060 \left(\frac{x_{mt}}{c} \right)^3 = 0$$

This is readily solved using Newton's method to yield

$$\frac{x_{mt}}{c} = 0.2998279$$

Using this in Eq. (4.1.31), we have

$$\frac{d^2t}{dx^2} = -7.533558 \frac{t_m}{c^2}$$

Using this result in Eq. (4.1.14), the radius of curvature at the point of maximum thickness is

$$r_c(x_{mt}) = \left[4 + \left(\frac{dt}{dx} \right)^2 \right]^{3/2} / 4 \left| \frac{d^2t}{dx^2} \right| = 2 / \left| -7.533558 \frac{t_m}{c^2} \right| = 0.2654788 \frac{c^2}{t_m}$$

or

$$\frac{r_{mt}}{c} = 0.2654788 \frac{c}{t_m}$$

From Eq. (4.1.29), the trailing edge thickness is

$$t(c) = (2.969 - 1.260 - 3.516 + 2.843 - 1.015) t_m$$

or

$$\frac{t_{te}}{c} = 0.021 \frac{t_m}{c}$$

Using Eq. (4.1.30) in Eq. (4.1.27), the trailing edge angle is given by

$$\begin{aligned} \theta_{te} &= 2 \tan^{-1} \left(-\frac{1}{2} \frac{dt}{dx}(c) \right) \\ &= 2 \tan^{-1} \left\{ -\frac{1}{2} \frac{t_m}{c} \left[\frac{2.969}{2} - 1.260 - 2(3.516) + 3(2.843) - 4(1.015) \right] \right\} \end{aligned}$$

or

$$\theta_{te} = 2 \tan^{-1} \left(1.16925 \frac{t_m}{c} \right)$$

For the NACA 0012

$$\frac{r_{le}}{c} = \frac{2.969^2}{8} \left(\frac{t_m}{c} \right)^2 = \frac{2.969^2}{8} (0.12)^2 = 0.015867$$

$$\frac{r_{mt}}{c} = 0.2654788 \frac{c}{t_m} = 0.2654788 \frac{1}{0.12} = 2.2123$$

$$\frac{t_{te}}{c} = 0.021 \frac{t_m}{c} = 0.021 (0.12) = 0.00252$$

$$\theta_{te} = 2 \tan^{-1} \left(1.16925 \frac{t_m}{c} \right) = 2 \tan^{-1} [1.16925 (0.12)] = 15.974^\circ$$